

Sunil Tiwari

Salt Lake City, UT | (318) 680-6123 | bishwast77@gmail.com
linkedin.com/in/suniltiwariacyber | github.com/bishwast | suniltiwari.me

CORE COMPETENCIES & TECHNICAL SKILLS

AI/ML & Orchestration: Large Language Models (LLMs), RAG Architectures, Prompt Engineering, LangChain, CrewAI, Llama 3.2, Ollama, Vector Retrieval **Backend & Cloud Systems:** Python (FastAPI), Multi-Step Tool-Using Agents, Azure, Docker Containerization, NVIDIA DGX (ARM64), PowerShell, Terraform **System Design & Engineering:** Agile/Scrum Methodologies, Project Lifecycle Management, Workflow Automation, SIEM (Splunk/Wazuh), Risk Mitigation, Change Management

AI ENGINEERING RESEARCH & PROJECTS

Hybrid-LLM Autonomous Agentic System (Lead Project Architect) December 2025 – Present

- Served as Project Lead architecting an end-to-end Hybrid-LLM orchestration system using Python and FastAPI, managing project milestones, delivery timelines, and overall scope.
- Coordinated the integration of isolated Docker honeypots with a custom FastAPI backend, engineering regex-driven Python telemetry bridges to dynamically extract execution data.
- Managed the execution of a deterministic Active Defense Governance Layer in Python to intercept AI-generated actuation commands, enforcing strict infrastructure IP whitelisting.
- Directed the porting of the agentic engine (CrewAI/LangChain) to defend a simulated multi-tenant Web Application stack, reconfiguring RAG playbooks to triage web-specific vulnerabilities.
- Achieved autonomous threat containment by programmatic timeline delivery of dynamic iptables drop rules against validated malicious endpoints.

Secure LLMOps Gateway / Production AI Proxy (Implementation Lead) May 2026

- Directed the engineering and deployment of a production-grade stateless security proxy in Python (FastAPI) designed to intercept, sanitize, and audit Generative AI prompt payloads.
- Managed the design of a high-precision Data Loss Prevention (DLP) engine to auto-redact PII, ensuring strict organizational compliance with PCI-DSS and SOC 2 data-in-motion standards.
- Coordinated the override of LLM safety-refusal guardrails on redacted payloads by developing a custom context preservation layer, reducing false-positive blocks for local Llama 3.2 models.
- Implemented SIEM-ready JSON logging frameworks to track processing latency and transaction UUIDs while completely preventing raw-text PII leakage.
- Hardened project delivery by packaging the proxy inside a lightweight Docker container configured to run under a restricted non-root system user.

Independent Autonomous Security & AI Lab (Lead Researcher) May 2025 – Present

- Designed secure local LLM inference pipelines using FastAPI and Docker to ensure privacy-compliant AI deployments on enterprise hardware (NVIDIA DGX ARM64).
- Simulated network intrusion scenarios to research and validate automated AI-driven mitigation playbooks.

PROFESSIONAL EXPERIENCE

Operations & IT Project Coordinator January 2024 – Present *MyBrow Threading* Orem, UT

- Led the end-to-end delivery of a digital client intake and automated scheduling system, managing product requirements, software selection, and testing phases.
- Integrated no-code architectures (Cal.com and Tally.so) with business workflows to optimize scheduling constraints and reduce administrative overhead.

- Managed client escalations, triaged system misconfigurations, and served as the single point of contact for technical problem resolution.

IT Operations & Security Manager August 2024 – April 2025 *Karuwaa Express* Sandy, UT

- Directed the implementation planning and timeline execution of a Formal Risk Treatment Plan for high-risk assets and enforced Role-Based Access Control (RBAC).
- Managed mandatory Staff Security Awareness Training deliverables, achieving a 0% phishing click rate.
- Mitigated internal fraud through least privilege enforcement and cross-departmental security monitoring.

Patient Access Specialist (Seasonal Contract) May 2024 – August 2024 *Intermountain Health*
American Fork, UT

- Validated patient identity and secured highly sensitive Protected Health Information (PHI) under strict HIPAA and PCI-DSS compliance frameworks.
- Identified and escalated potential identity fraud anomalies during high-volume digital intake processes.

Partner Support Specialist September 2020 – September 2021 *Walmart eCommerce* Remote

- Managed complex enterprise workflows across Salesforce instances, coordinating the resolution of high-priority technical and partner issues.
- Drove business process optimization projects that resulted in a 20% verified increase in overall team delivery and output efficiency.

IT Support Help Desk January 2019 – May 2019 *Century Link* Monroe, LA

- Managed Active Directory for 100+ endpoints, enforcing GPOs and resolving Access Control List (ACL) violations.
- Provided remote systems support and integration troubleshooting for VPN and Citrix infrastructure using PowerShell.

EDUCATION & CERTIFICATIONS

Master of Science in Cybersecurity Management Expected July 2026 *University of Utah* Salt Lake City, UT
Specialization: Agentic AI in SecOps

Bachelor of Science in Information Technology December 2022 *Utah Valley University* Orem, UT
Network Security Focus

Certifications: CompTIA Certified Cybersecurity Analyst (CySA+) | Systems Security AI+ (SecAI+)